HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and executed test plans, procedures, and acceptance criteria derived from product requirements for enterprise systems.
- Configured test articles, instrumentation, data acquisition systems, and laboratory setups for complex engineering assignments.
- Collected, analyzed, and reported test data with meticulous attention to accuracy, traceability, and measurement uncertainty.
- Developed and maintained automated test scripts and data-processing workflows to streamline testing procedures.
- Documented anomalies, reproduced failures, and supported root cause investigations, verifying corrective actions with design teams.
- Supported functional, environmental, reliability, regression, compliance, and system integration testing across multidisciplinary teams.
- Collaborated with design, systems, manufacturing, quality, regulatory, service, and project teams to achieve product objectives.

SKILLS

Testing & QA – Test Plans, Test Procedures, Acceptance Criteria, Data Acquisition, Regression Testing, System Integration Testing, Reliability Testing, Compliance Testing, Environmental Testing, Failure Analysis, Root Cause Analysis, Corrective Actions

Programming Languages – Python, Embedded C

Data Analysis & Reporting – Data Analysis, Data Reporting, Measurement Uncertainty, Repeatability, Reproducibility

Core Engineering – Engineering Principles, Verification, Validation, Test Method Development, Laboratory Instrumentation, Engineering Measurement, Product Requirements, Design Quality, Technical Documentation, Issue Resolution

Collaboration & Process – Multidisciplinary Teams, Design Teams, Systems Engineering, Manufacturing, Quality Assurance, Regulatory Compliance, Project Management, Product Lifecycle Management, Configuration Management

PROJECTS

Automated Test Fixture Development for Industrial Sensors

- Designed and implemented a Python-based automation framework for testing industrial sensor modules, reducing manual test time by 40%.
- Integrated data acquisition systems and custom fixtures to collect real-time performance metrics, ensuring accurate measurement of sensor output under varied conditions.
- Developed objective, repeatable, and traceable test methods based on product requirements for validation and qualification.

Python Scripting for Test Data Generation and Analysis

- Engineered Python scripts to generate synthetic test data sets for edge-case scenarios, improving test coverage for system integration testing.
- Developed data processing workflows to analyze large volumes of test results, identifying trends and anomalies with statistical significance.
- Documented test execution results and supported root cause analysis for reported failures, contributing to faster issue resolution.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)